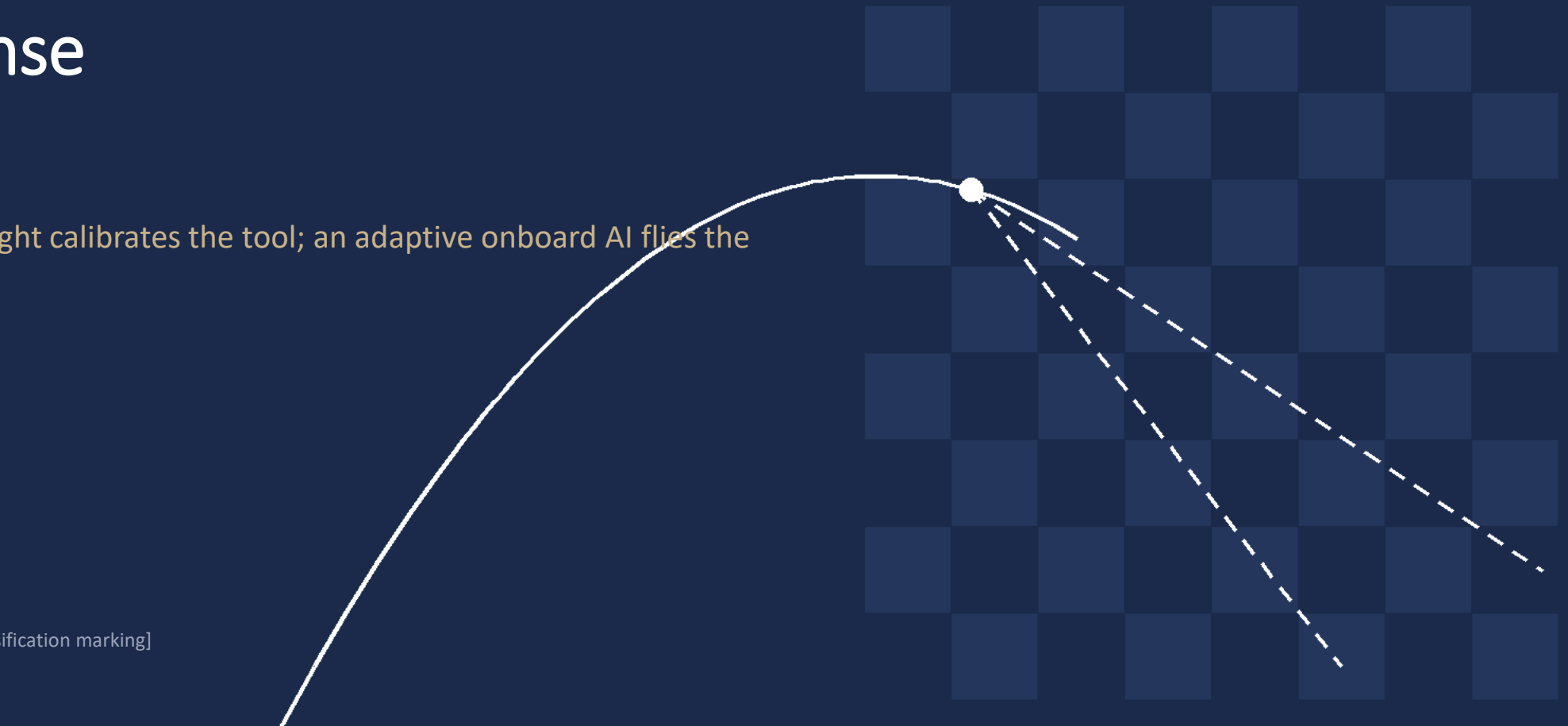
CHESS — OPUS AEROSPACE · CAID 2026

# AI-driven, flight-proven threat scenarios for air defense

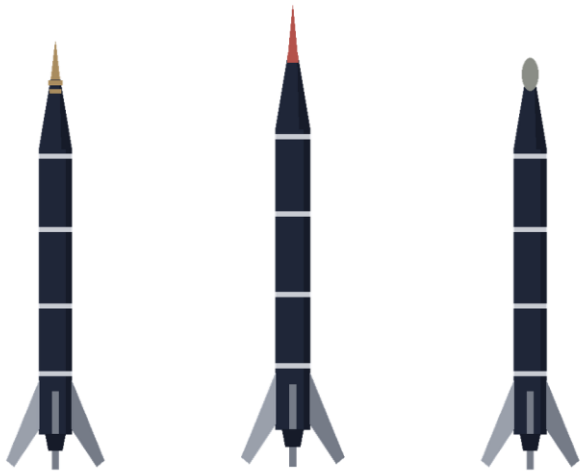
Each suborbital flight calibrates the tool; an adaptive onboard AI flies the threat.



[Presenter] · [Date] · [Classification marking]

# Why “CHESS” — decisions against the clock

The name is not about playing games: deciding under time pressure, from validated moves only



Queen King Pawn

*Every threat in a CHESS salvo plays a piece’s role — the doctrine comes in part 1.*

| | In chess | In CHESS |
|---|---|---|
| **Calibration** | Engine tuned on real games | The tool is calibrated on every flight |
| **Preparation** | Opening book before the game | Mission decision tree uplinked before the flight |
| **Decision** | Best move from the position | Onboard AI selects the next branch from sensor data |
| **Clock** | Each move played in time | Bounded compute (WCET) |

*One difference: CHESS only plays moves that were validated on the ground.*

# Agenda

Four questions, one learning loop

**1** **Use cases**

What do today's synthetic threats miss, and what does air defense need?

**2** **Technical approach**

What runs today, and how does each flight calibrate the tool?

**3** **On-board AI**

How does the threat adapt in flight while staying bounded?

**4** **Maturity & evaluation**

Where do we stand, and how will results be measured?

# Limits of synthetic threats

Today's threats rest on models, not flights

| | Synthetic threats today | Flight-proven threats (CHESS) |
|---|---|---|
| **Trajectory** | Modelled from assumed aerodynamics and propulsion | 6-DOF reconstruction of real flights, ground and onboard sensors |
| **Signatures** | RCS and IR predicted, rarely validated | Measured on a jet-like threat profile |
| **Uncertainty** | Error bars unknown | Dispersion statistics, quantified residual error |
| **Behaviour** | Scripted, cannot react | Onboard AI adapts in flight, within a validated decision tree |

*This is the gap CHESS is built to close: threat scenarios whose physics carry known error bars.*

# Value for air defense

Every scenario is anchored in measured flight data, not assumptions

### C2 evaluation bench

Assess C2 chains against raids whose trajectories and signatures were measured.

### Operator training

Train crews on threat behaviour that has actually flown, including in-flight adaptation.

### GBAD planning

Plan ground-based air defense on doctrinal and achievable scenarios.

A live exercise: **months of preparation for a single pass.** In CHESS: **a scenario replays in minutes, without engaging any real asset.**

*Your operational needs feed directly into the definition of the next flight.*

# The CHESS doctrine — every threat plays a role

The salvo is a plan, not a script

### King — the real threat

The real trajectory, complete. The one that must be detected early enough to engage.



### Queen — the premium decoy

Replays the King's flight, rotated — a larger signature, built to steal the track.



### Pawns — saturation

Own physics, short burn, launched in numbers — to drown the King in noise.



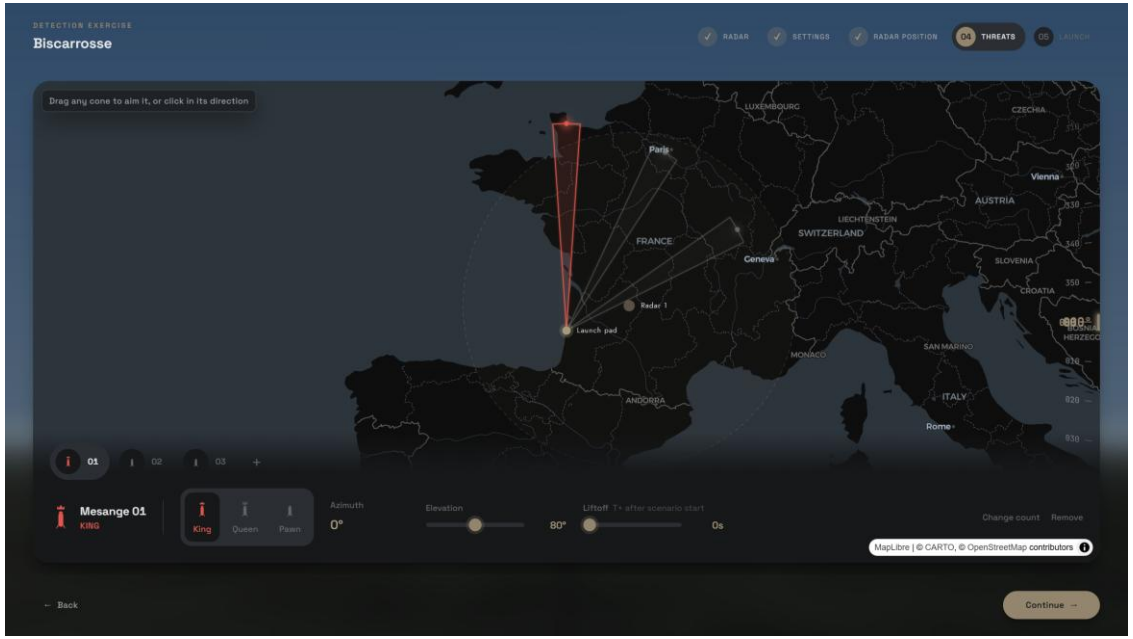
*The salvo plays against the operator: a composition designed to exploit the weaknesses of the defense layout. Crews train against an adversary — not against a script.*

# What runs today — the simulator

## A complete exercise flow, operational now




**Launch-site selection.** Six real European bases — site weather and elevation feed the simulation.

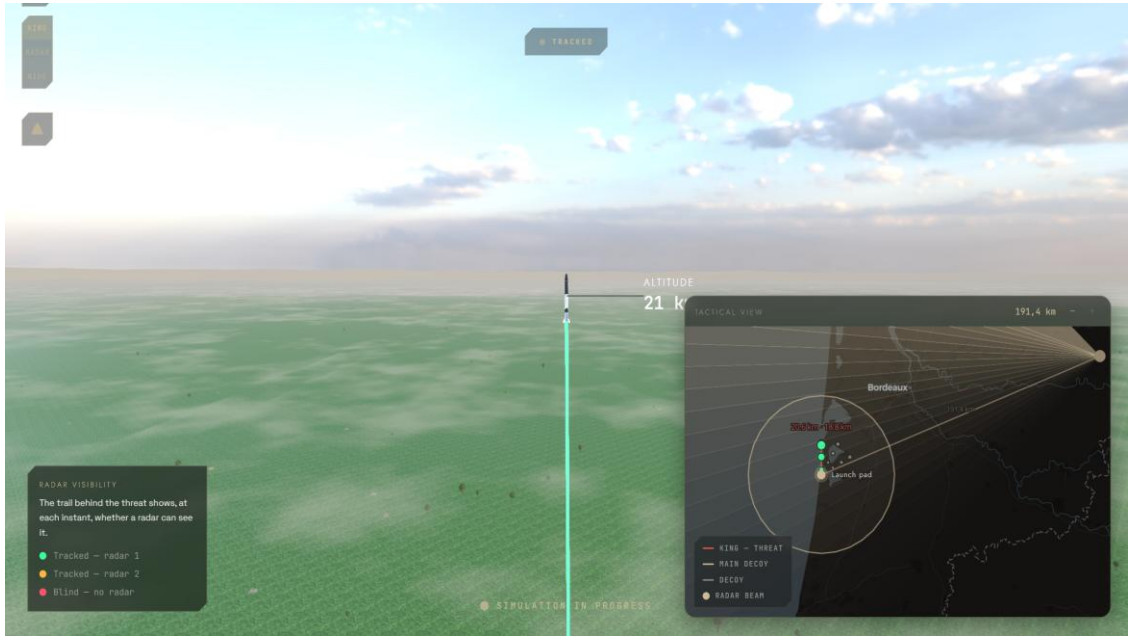



**Salvo composition.** 1-5 threats, azimuth and elevation per threat, roles assigned by the doctrine.

*Running today: the physics simulation core and the full exercise flow. The AI layers are being built on top of it.*

# What runs today — flight & debrief

Physics-based detection, not a distance check




**Real-time 3D flight.** Tactical map, radar beam, live detection status of the King.

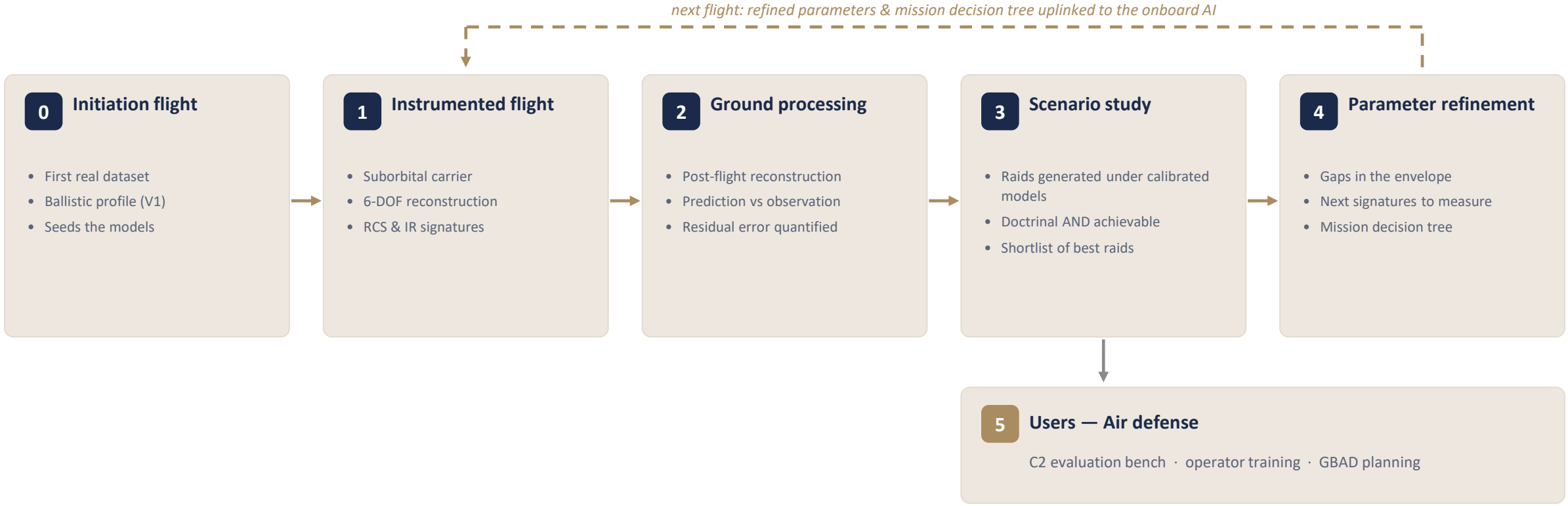



**Post-flight debrief.** Coverage volume, vertical cut, network verdict, per-radar analysis.

Radar horizon (4/3 refraction) · radar equation (4th-root RCS) · aspect RCS: ~0.08 m² nose-on vs ~1.2 m² broadside · ceiling & scan rate

# The iterative loop — one cycle per flight

Fly, measure, calibrate, refine




*Each flight calibrates the tool: models sharpen, scenarios tighten, the decision tree improves.*

# Calibrated tool, adaptive AI

Calibrated on the ground, adaptive in flight

| *Ground · calibrated tool* | | *Onboard · adaptive AI* |
|---|---|---|
| Models calibrated on reconstructed flight data | | Receives the refined tree before each flight |
| Raid generation constrained by the calibrated models | *uplink* | Processes onboard sensor data in real time |
| Optimization shortlists doctrinal and achievable raids | | Adapts its choices to what it perceives, like a chess move |
| Outputs the mission decision tree | | Adapts the trajectory within bounded compute (WCET) |

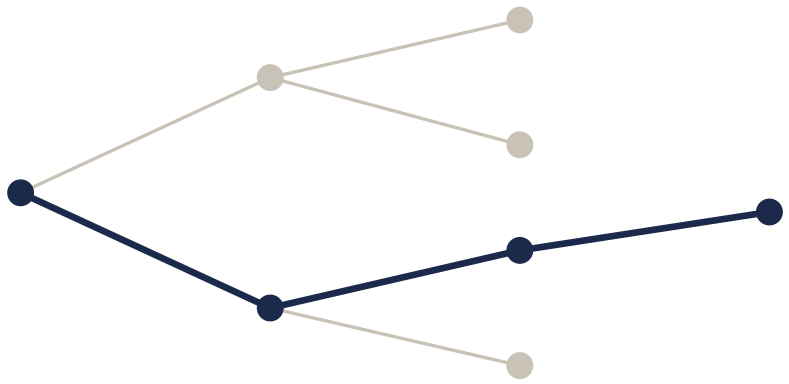
**Flights calibrate the tool; the onboard AI adapts in flight.**

# On-board AI

The threat adapts in flight, like a chess player

**Onboard sensors** → **Sensor processing** → **'Chess' branch selection** → **Trajectory adaptation**

*Bounded compute (WCET): predictable worst-case timing at every step*



## Adaptive, never unbounded

Each move is chosen in flight from the uplinked mission decision tree. Every branch was defined and refined on the ground.

# Where exactly is the AI?

Three components, one deliberate design choice

### Learned layer — scenario generation

Candidate raids are generated and scored under models calibrated on flight data. The learned layer constrains what is doctrinal AND achievable.

IN DEV

### Onboard perception

Real-time processing of onboard sensor data, to recognise the situation among the prepared branches of the tree.

IN DEV

### Bounded decision policy

The uplinked mission decision tree. Every branch validated on the ground, every step within proven time bounds — a qualifiability choice, not a limitation.

IN DEV

*Frugal, bounded, qualifiable — by design.*

*We chose a bounded policy before building the AI: in this domain, an explainable decision beats an impressive one.*

# Where we stand today

An early-stage project — stated plainly

**RUNNING TODAY**

- Physics simulation core
- Full exercise flow: sites, radars, salvo, 3D flight, debrief
- Detection physics with per-radar verdicts

**IN DEVELOPMENT**

- Configuration advisor (search over the simulator)
- Raid generation & optimization — the learned layer
- Onboard policy prototype (bounded branch selection)

**PLANNED**

- First instrumented flights
- First calibration cycle on measured data
- Adversary evaluation campaign vs scripted baseline

*No claimed results, no demo magic: this page is the contract we intend to be measured against.*

# How we will measure success

The evaluation protocol exists before the AI

### Adversary value

Baseline: scripted salvos.
Metrics: blind-spot discovery rate, reduction of defender lead time at equal raid budget.

### Calibration quality

Residual error between prediction and observation, quantified after each flight — published per cycle.

### Bounded execution

Measured worst-case execution time on the target compute — every branch, every step, every time.

*Results will be measured, not claimed.*

# Hard problems we are working on

Named now, so they don't surprise us later

### Sim-to-real gap

How far does a model calibrated on N flights remain valid outside the measured envelope? Extrapolation must carry explicit error bars.

### In-flight signature measurement

Instrumenting RCS & IR on a suborbital carrier is hard: sensor choice, geometry, and reconciliation with ground measurements are open work.

### Real-time guarantees

Keeping every decision step within proven worst-case bounds (WCET) on embedded hardware, as the tree grows richer.

*If you have fought one of these problems, we want to talk to you.*

# Conclusion

Closing the loop between flight and air defense

**01**

### Measured, not modelled

Reconstructed trajectories and measured signatures, with known error bars.

**02**

### Calibrated with every flight

Each flight calibrates the tool and closes a gap in the envelope.

**03**

### Adaptive, yet bounded

The onboard AI adapts in flight within a validated decision tree and bounded compute.

*What we are looking for: operational feedback on the use cases · experimentation partners · flight-test opportunities*

[Contact name] · [email]



# Thank you

*Questions?*

[Contact name] · [email] · [phone]

# Annex · Roadmap

Next developments of the tool

| | |
|---|---|
| **Phase 0**<br>*[Q_ 20__]* | **Initiation flight**<br>Ballistic profile (V1): first real dataset, seeds the models |
| **Phase 1**<br>*[Q_ 20__]* | **First instrumented flights**<br>6-DOF reconstruction, RCS & IR signatures, dispersion: first calibration of the tool |
| **Phase 2**<br>*[Q_ 20__]* | **Scenario generation for users**<br>Raids generated under calibrated models, shortlists for C2 bench, training and GBAD |
| **Phase 3**<br>*[Q_ 20__]* | **Adaptive onboard AI in flight**<br>Real-time branch selection within the uplinked tree, bounded compute (WCET) |
| **Phase 4**<br>*[Q_ 20__]* | **Operational loop**<br>One cycle per flight, operational needs drive each next flight definition |